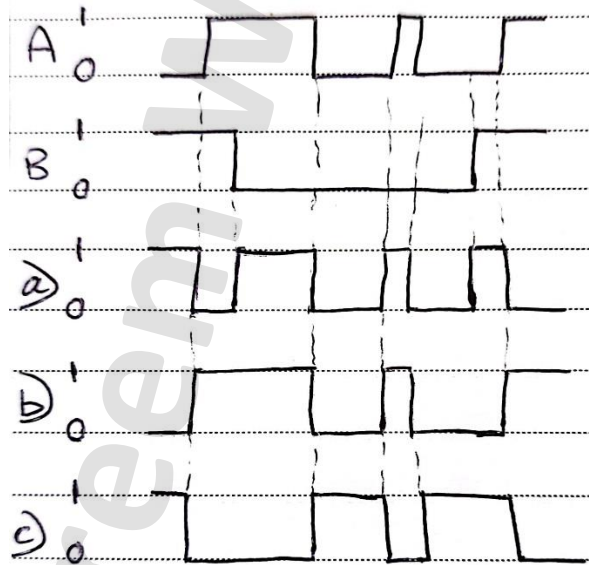


1)

```
module ct1(A, out);  
  input [3:0] A;  
  output out;  
  
  assign out = (A > 4'b0010 && A < 4'b1000)? 1: 0;  
  
endmodule
```

2)



CS Scanned with CamScanner

3) $F = 1$ when C is HIGH, B is HIGH and A is LOW

```
module ct3(A, B, C, out);  
    input A, B, C;  
    output F;  
  
    wire XOR_out, XNOR_out;  
  
    assign XOR_out = A ^ B;  
    assign XNOR_out = B ~^ C;  
    assign F = C & XOR_out & XNOR_out;  
  
endmodule
```

4) Using 3-input NOR gate

5)

```
module ct7(A, B, Ainvert, Binvert, Operation, CarryIn, Carryout, Result);
    input A, B, Ainvert, Binvert, CarryIn;
    input [1:0] Operation;
    output Carryout, Result;

    wire Amux_out, Bmux_out, AND_out, OR_out, adder_out;

    assign Amux_out = (Ainvert == 1)? ~A : A;
    assign Bmux_out = (Binvert == 1)? ~B : B;

    assign AND_out = Amux_out & Bmux_out;
    assign OR_out = Amux_out | Bmux_out;
    assign {Carryout, adder_out} = Amux_out + Bmux_out + CarryIn;

    assign Result = (Operation == 0)? AND_out: (Operation == 1)? OR_out: adder_out;
endmodule
```